

GEN-AI STUDIO: E-COMMERCE PRODUCT PHOTOGRAPHY VIA AN END-TO-END MLOPS PIPELINE

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Project Task

The Problem

- Generative AI models perform well in research settings.
- Deployment introduces GPU memory limits, latency, and environment issues.
- Small e-commerce businesses cannot afford professional product photography.

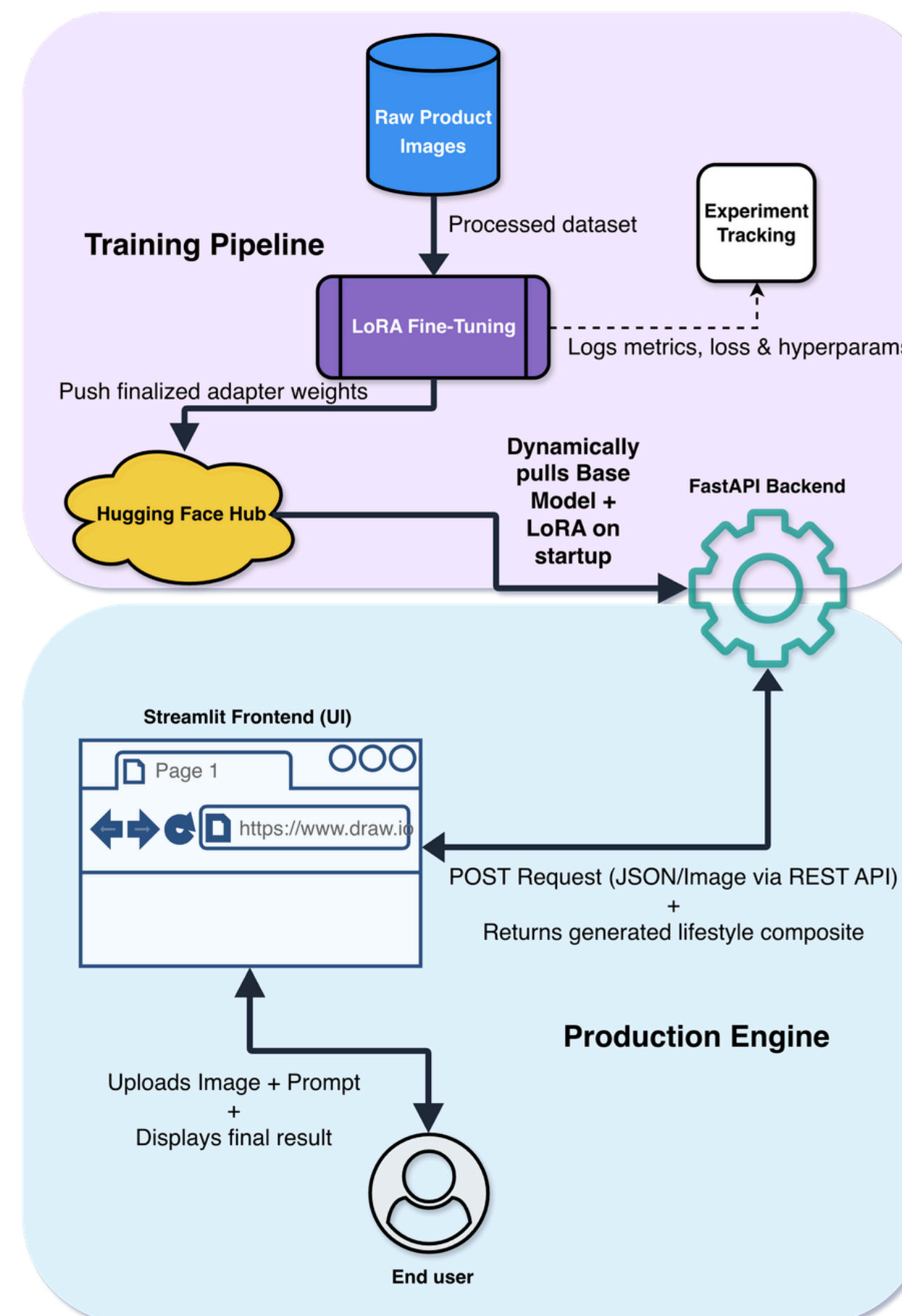
Solution

This project builds a **containerized Generative AI pipeline** that converts simple product photos into **professional lifestyle images**. Using **LoRA to adapt a pre-trained inpainting model**, the system separates the heavy backend from the lightweight UI for stable and scalable deployment.

Deployment Stages

- Model Adaptation** – Fine-tune the **Stable Diffusion Inpainting** model using **LoRA** so it preserves the product while generating new backgrounds.
- Model Versioning** – Store the trained adapter weights on the **Hugging Face Hub** as a centralized model registry.
- Backend API Development** – Build a **FastAPI backend** with a /generate endpoint to process prompts and generate images.
- Interactive Frontend** – Create a **Streamlit UI** where users upload product images, enter prompts, and view results
- Containerized Deployment** – Use **Docker and docker-compose** to run the backend and frontend as separate containers for reliable deployment.

Build a Generative AI system that turns simple product photos into professional lifestyle images. Users upload a product image and describe a scene with a text prompt, and the system generates a realistic background around the product while keeping it intact.



Libraries / APIs to be Used

HuggingFace → Model loading & LoRA fine-tuning
Docker → Containerized deployment
FastAPI → Backend API
Streamlit → Interactive UI
MLflow/WandB → Experiment tracking
PyTorch → Core ML framework

BEFORE-AFTER



Datasets and Models to be Used

Dataset: amazon-shoe-images (HuggingFace) – product images on plain backgrounds.

Model: Stable Diffusion Inpainting – preserves product while generating new scene from prompt.

Product Image + Prompt

Stable Diffusion Inpainting

Lifestyle Product Photo